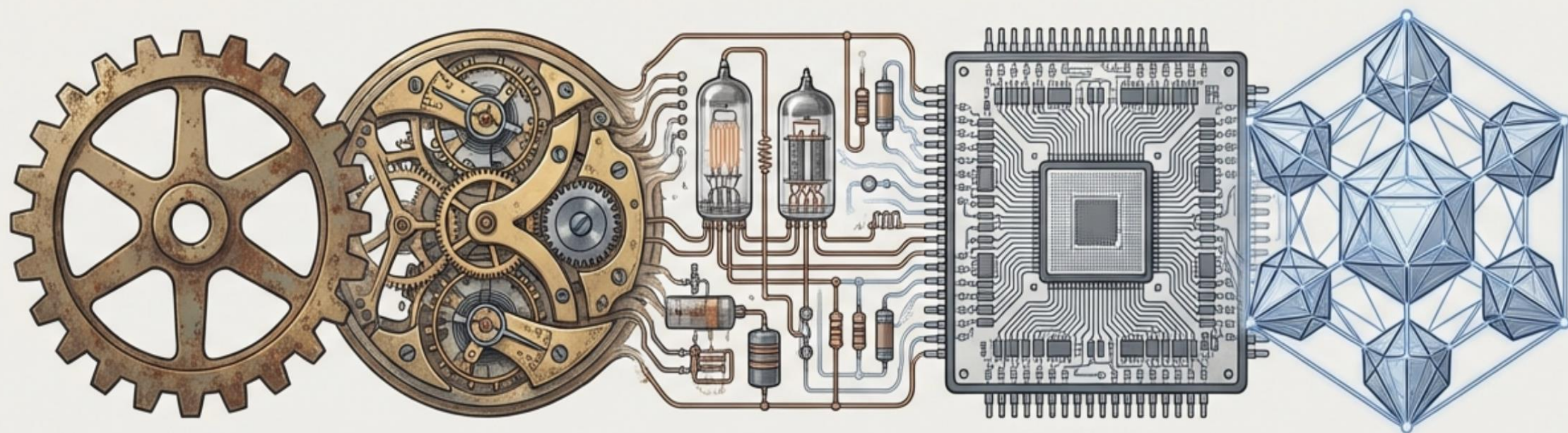




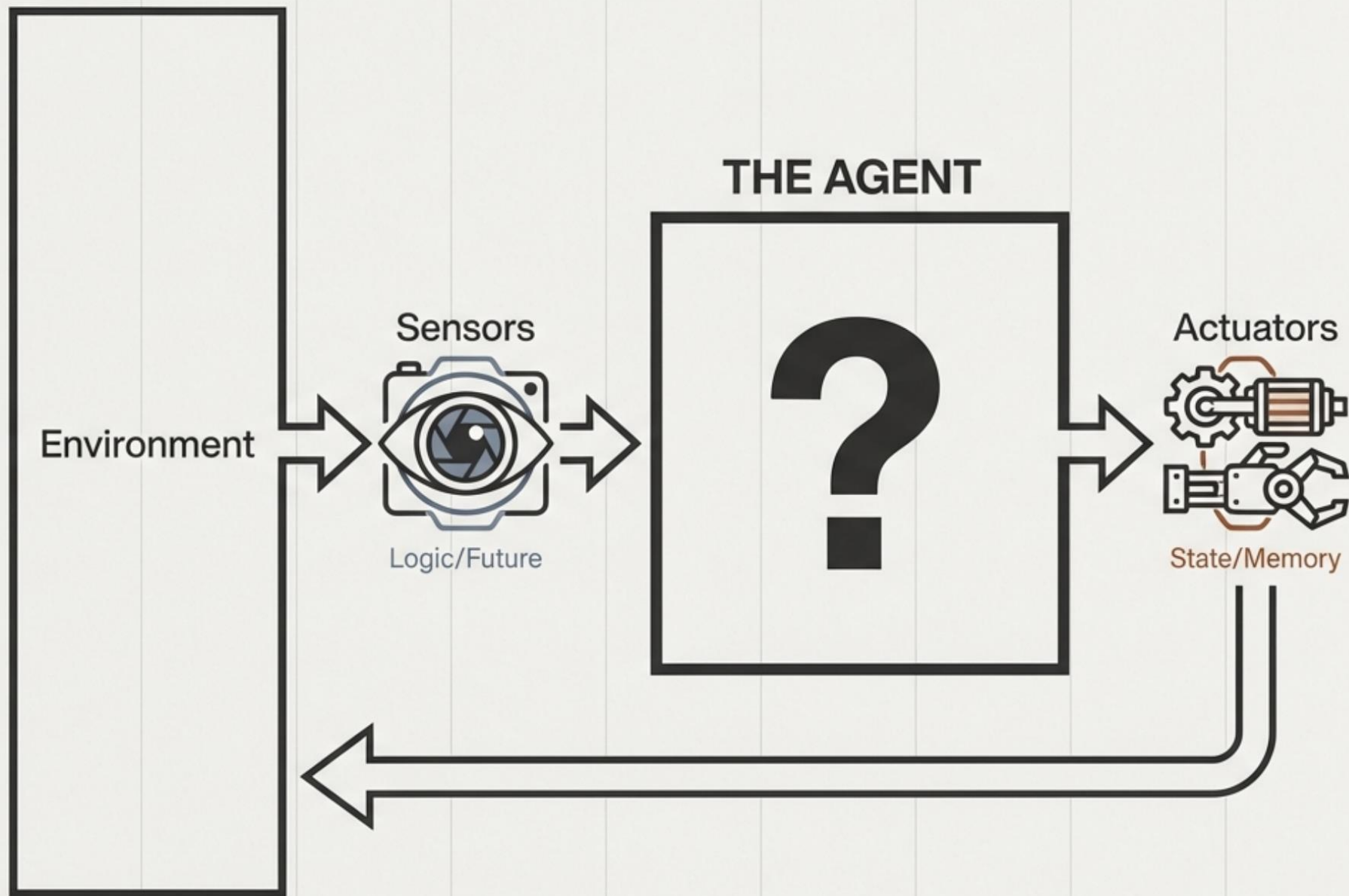
From Reflex to Reason

The Architectures and Evolution of Artificial Intelligence Agents

The Cycle of Perception and Action

At its core, an **Agent** is any entity that perceives its environment through sensors and acts upon that environment through actuators. The complexity of an agent is defined by how it bridges the gap between the input (Percept) and the output (Action).

The Goal: To open the "Black Box" and trace the evolution of the decision-making logic inside.
Domaine Text Regular

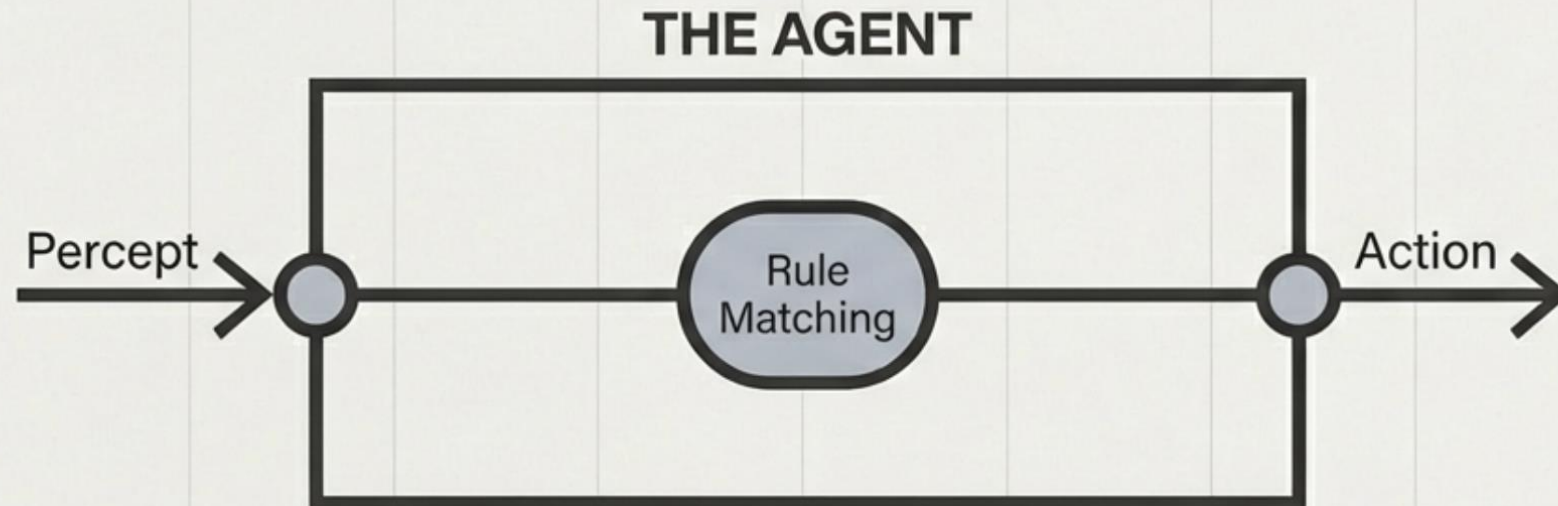


Level 1: The Simple Reflex Agent

The Immediate Reactor

This agent operates on a "Condition-Action" basis. It acts only on the current percept. It has no history (no memory of the past) and no foresight (no reasoning about the future).

IF condition THEN action



Example: The Thermostat

Details

Percept: Current Temperature.

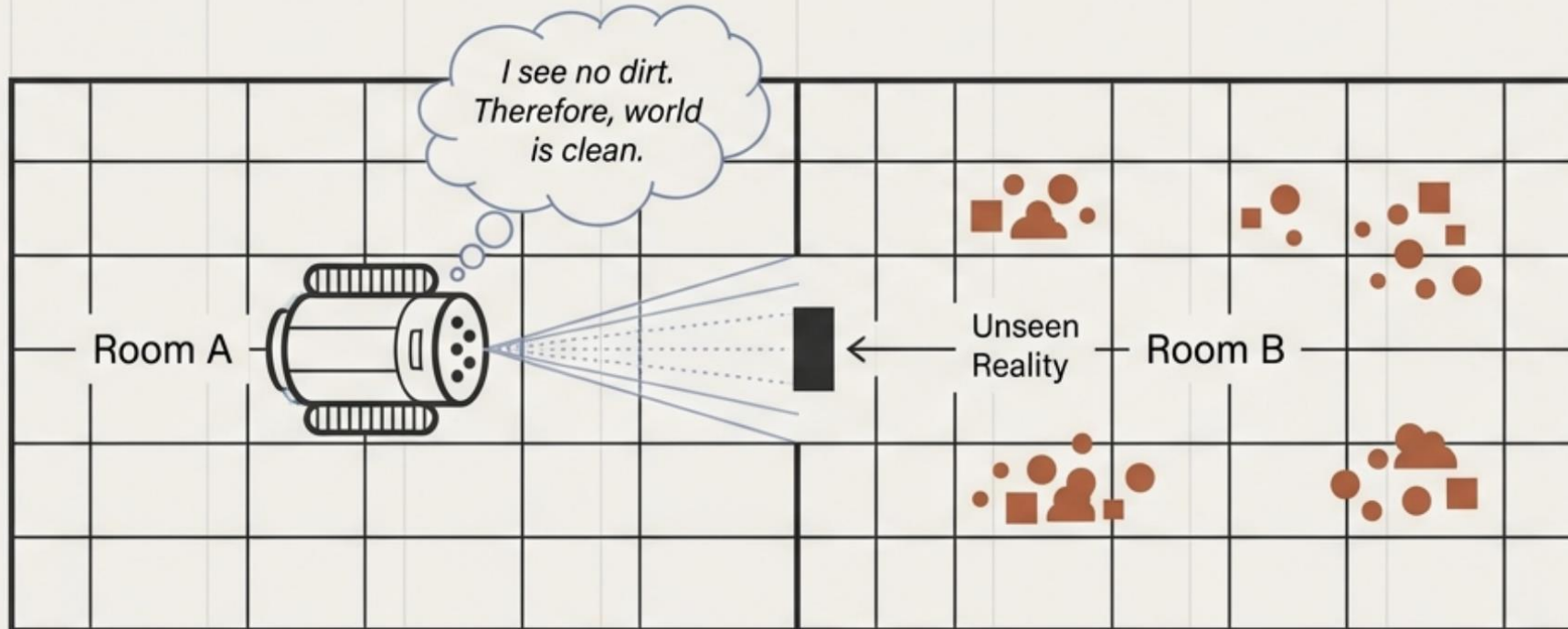
Rule: If $< 20^{\circ}\text{C}$, Heater ON.

Note: It does not care if it was freezing yesterday. It only knows "now".

The Failure of Blindness

Simple reflex agents only function in fully observable environments. If the sensors cannot see the entire state of the world, a simple rule leads to irrationality.

Blindness to the past leads to errors in the present.



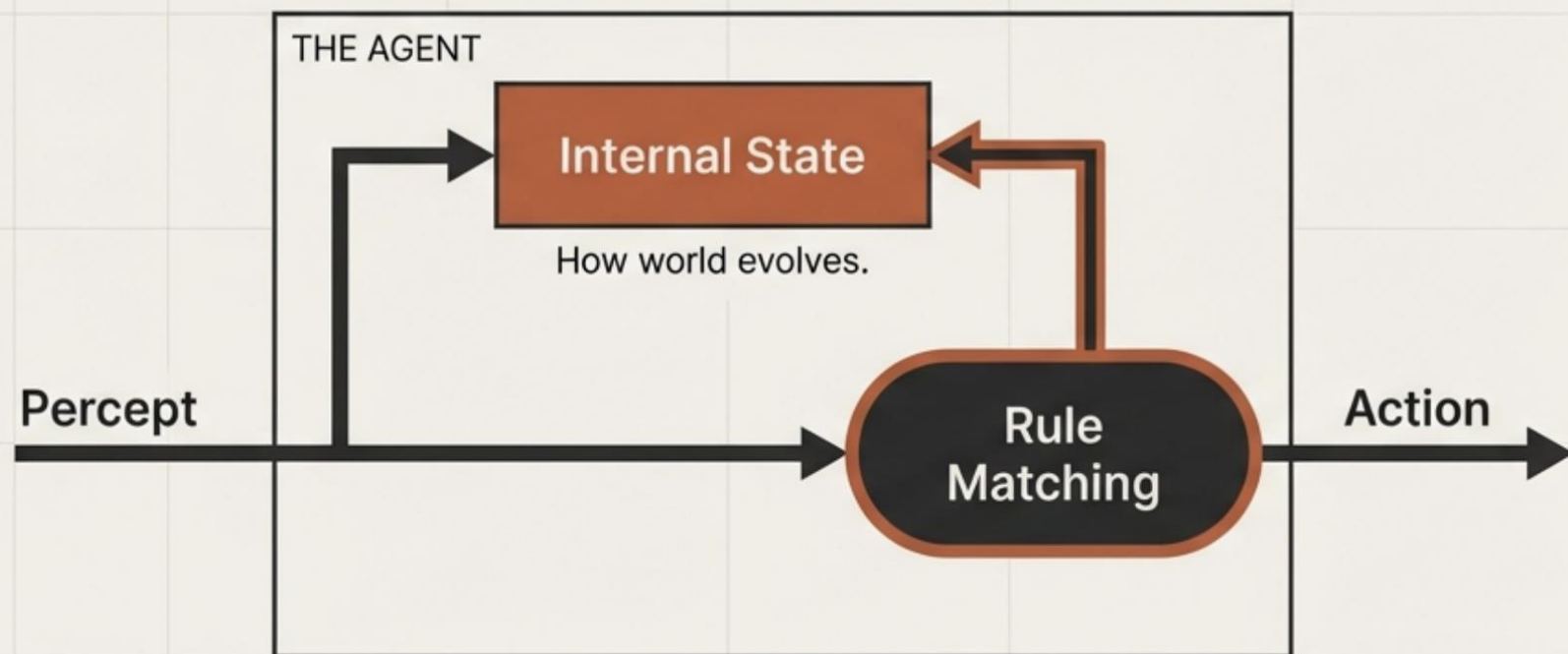
To survive partial observability, an agent needs Memory.

Level 2: Reflex Agents with State

The Agent with Memory

These agents maintain an **Internal State**—a memory model that tracks aspects of the world not currently visible. They combine current percepts with history.

Current Percept + Internal State → Rule Matching → Action



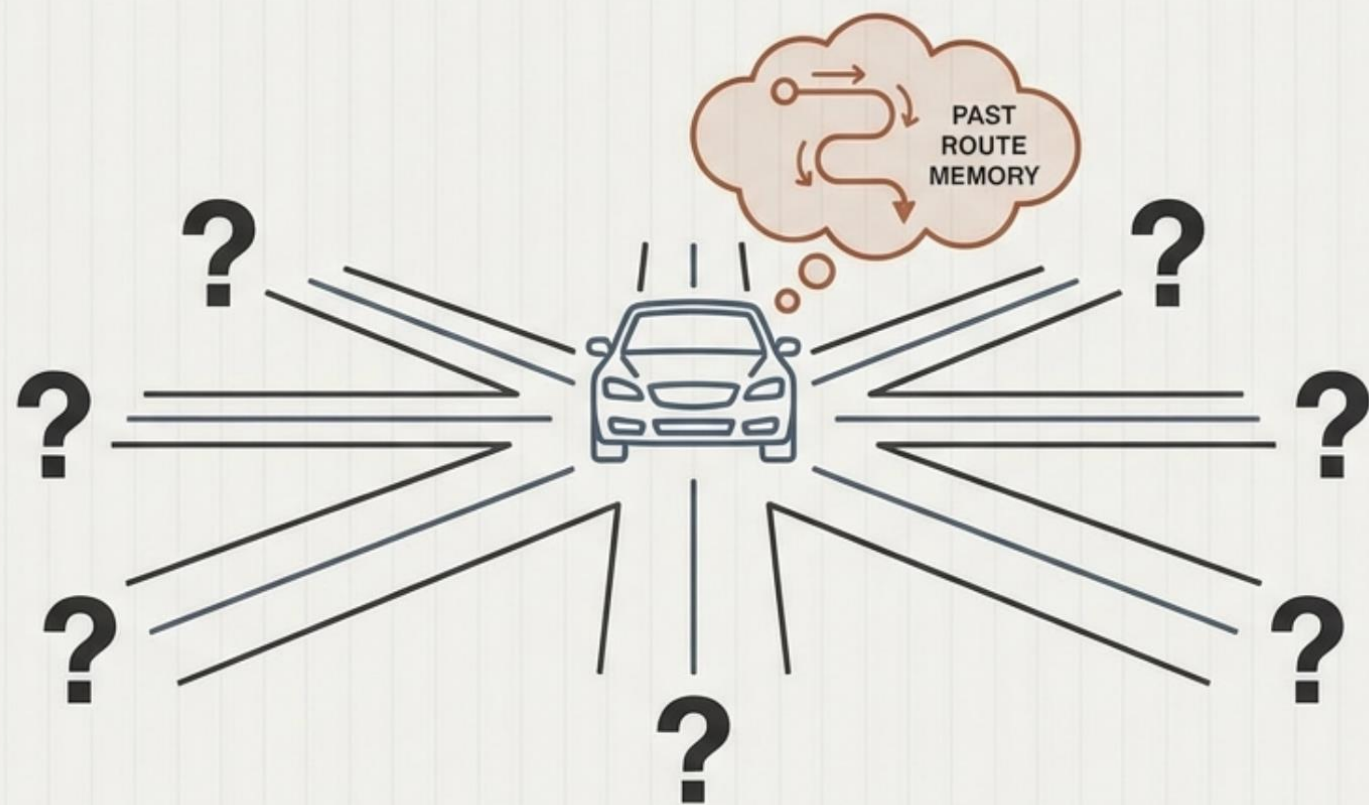
Example: The Blind Spot

Scenario: A passing car disappears into a blind spot.

The Fix: The **Internal State** "remembers" the car's velocity and position even when sensors see nothing, preventing a collision.

The Limits of Reaction

A Reflex Agent with State knows the history of the world, but it is still just following pre-written rules. It doesn't know what it is trying to achieve. Knowing the room is dirty (State) is different from *wanting* the room to be clean (Goal).



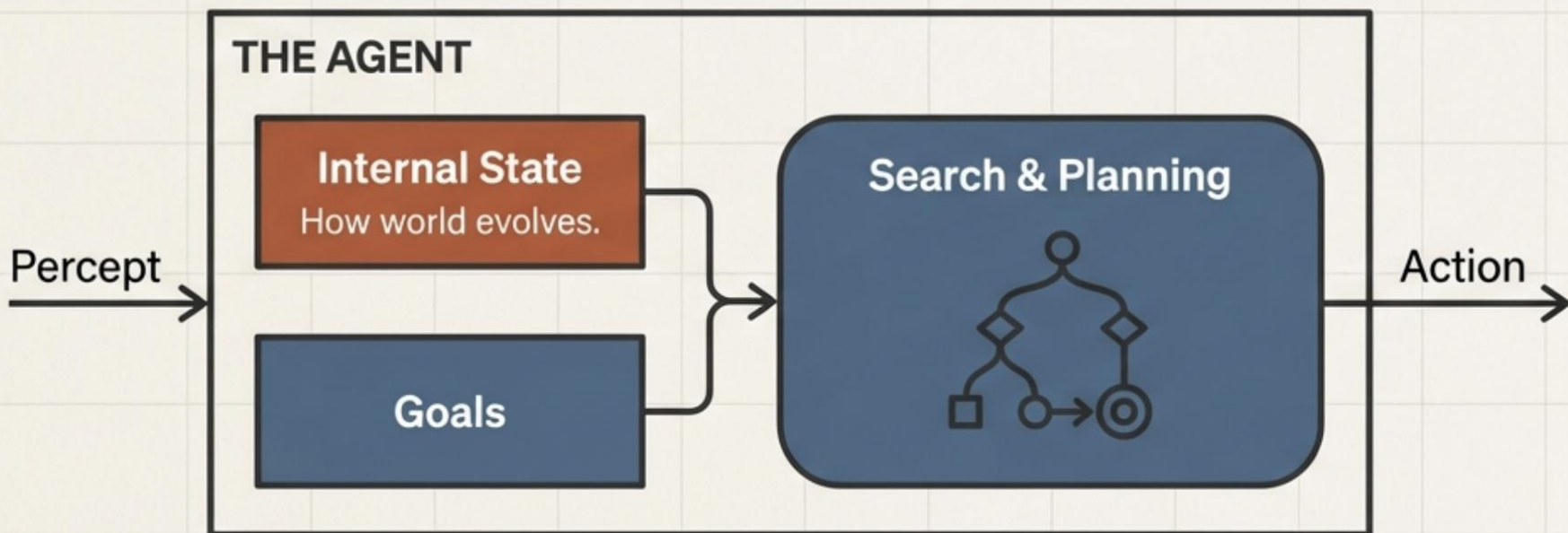
To stop reacting and start planning, we need Goals.

Level 3: Goal-Based Agents

The Planner

This agent searches and plans. It asks, "Will this action help me achieve my goal?" It considers future states and selects the action sequence that leads to the desired outcome.

Algorithm: State $\rightarrow$ Goals $\rightarrow$ Search/Planning $\rightarrow$ Action



Example: Chess & GPS

Process: The agent simulates future moves. It doesn't blindly react to a pawn move; it *plans* 5 steps ahead to trap the King.

The 'Any Goal Will Do' Fallacy

A Goal-Based agent treats all successful outcomes as equal. It cannot distinguish between a highly efficient solution and a barely passable one.



To a Goal-Based agent, these solutions are identical.

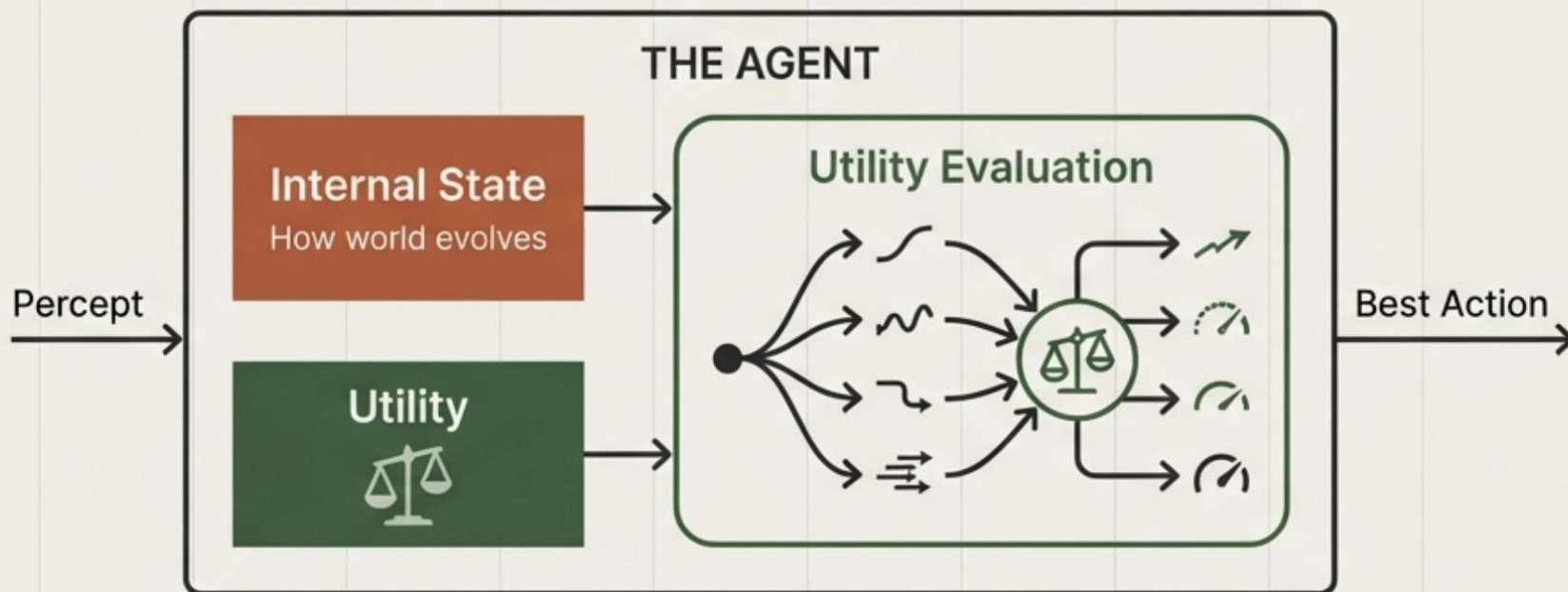
To distinguish between 'good' and 'best', we need Utility.

Level 4: Utility-Based Agents

The Optimizer

These agents ask, “How *good* is this outcome?” They replace binary goals with a **Utility Function**—an internal measure of performance, happiness, or preference.

State → Outcomes → Utility Evaluation → Best Action



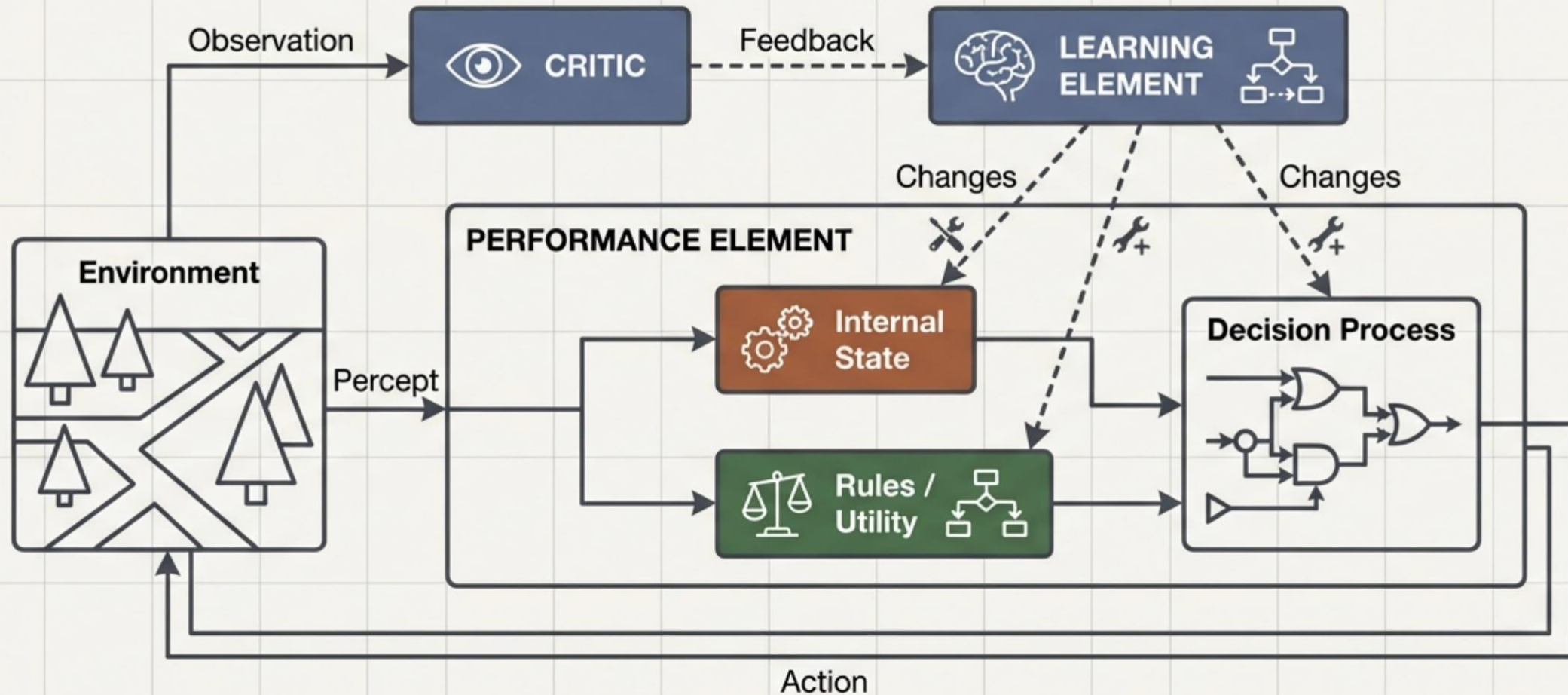
Example: The Balancing Act

Scenario: Autonomous Driving.

The Trade-off: The car must weigh Safety vs. Speed vs. Comfort. It calculates the “Maximum Expected Utility” (e.g., braking gently instead of slamming stops).

The Capstone: Learning Agents

Static vs. Dynamic. All previous agents can be hard-coded. A Learning Agent evolves its own rules and utility functions over time.



The Spectrum of Intelligence

Agent Type	Memory (State)	Goals (Future)	Utility (Preference)	Best Use Case
Simple Reflex	✗	✗	✗	Thermostats, Fully Observable tasks.
Reflex with State	✓	✗	✗	Basic robots, Tracking motion.
Goal-Based	✓	✓	✗	GPS, Chess, Games.
Utility-Based	✓	✓	✓	Financial trading, Recommendation Engines.

Designing for the Environment

As environments become less observable and more complex, the computational cost of the agent rises. The history of AI architecture is the history of building internal machinery to compensate for the uncertainty of the external world.

